Project Phase 2: Project Progress Report

Predicting FIFA World Cup 2026 Outcomes Using SQL and Machine Learning in Python

Team members

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1. Problem Statement

Predicting outcomes in international football tournaments requires understanding how relative team strength, contextual factors, and historical performance patterns jointly influence match results. Based on initial exploratory data analysis, this project focuses on modeling match-level outcomes using pre-match information, particularly team strength indicators and venue context.

Preliminary analysis indicates that relative ELO rating difference between competing teams is strongly associated with match outcomes, while venue type (neutral vs non-neutral) exhibits a measurable home advantage effect. Tournament type appears to have comparatively weaker influence once strength and venue are considered.

Accordingly, this project aims to develop a predictive framework to estimate match outcomes for international football competitions, with emphasis on applying the model to simulate potential outcomes for the FIFA World Cup 2026. The modeling approach will rely exclusively on pre-match variables to prevent data leakage and ensure realistic predictive performance.

In addition to classification of match winners, the project will explore goal-scoring dynamics. Exploratory analysis reveals overdispersion in total goals scored, suggesting that count-based modeling approaches such as Poisson or Negative Binomial regression may be appropriate for estimating scoring distributions.

By integrating competitive strength measures, contextual venue information, and statistically sound modeling techniques, this project seeks to construct an interpretable and empirically grounded predictive framework for international football match outcomes.

2. Data Preprocessing

2.1 Data Cleaning

The dataset was first examined for missing values, duplicate records, and data type inconsistencies. Missing values were assessed using summary functions, and no critical missing fields were found in the key analytical variables (ELO ratings, match outcomes, venue type). Duplicate records were checked and removed where necessary to ensure data integrity.

Data types were verified to confirm that numeric variables (e.g., ELO ratings, goals scored) were stored as integers or floats, and categorical variables (e.g., tournament type, venue indicator) were correctly formatted. Binary variables such as home_win and neutral were encoded as 0/1 to facilitate statistical analysis.

Outlier detection was conducted on the total_goals variable using descriptive statistics and IQR-based thresholds. While a small number of matches exhibited unusually high goal totals (greater than 12 goals), these cases were retained because they represent legitimate historical outcomes. Instead of removing them, their impact was considered in model selection (e.g., accounting for overdispersion in count modeling).

Data validation steps included checking logical consistency, such as ensuring that home_win aligned correctly with score differences and that derived variables matched original match results.

2.2 Data Integration

The dataset used in this project was constructed from structured match-level data that included team ratings, match outcomes, and contextual variables. Where applicable, datasets were merged using consistent match identifiers and aligned schemas. Key consistency checks were performed to ensure that ELO ratings corresponded correctly to the appropriate teams and match records.

2.3 Data Normalization and Feature Engineering

Several new features were engineered to enhance interpretability and predictive relevance.

The variable elo_diff was created as the difference between home and away ELO ratings:

$$elo_diff = home_elo - away_elo$$

This variable captures relative team strength and reduces multicollinearity that would arise from including both raw ELO values. Since match outcomes depend on relative strength, elo_diff was selected as the primary strength-based predictor.

The variable goal_diff was computed as the difference between home and away scores. Although useful for descriptive analysis, it was excluded from predictive modeling because it is derived from post-match information and directly determines the outcome variable, which would introduce data leakage.

Categorical variables such as tournament_type were retained in categorical form during exploratory analysis. Prior to modeling, these variables will be encoded using one-hot encoding to ensure compatibility with regression-based methods.

Feature scaling was not applied at this stage. Logistic regression does not require scaled predictors for correctness, and the main engineered variable (elo_diff) is already interpretable in its natural scale. If regularized or distance-based models are introduced in later phases, scaling will be considered.

3. Preliminary Results of Exploratory Data Analysis (EDA)

3.1 Statistical Analysis

Descriptive Statistics

The dataset contains 49,071 international football matches. The average total goals per match is 2.94, with a standard deviation of 2.09. The median number of goals is 3, indicating a moderately symmetric central distribution with right-skewed tail behavior.

Variance (4.37) exceeds the mean (2.94), suggesting overdispersion relative to a Poisson distribution. This indicates that goal counts may be better modeled using a Negative Binomial framework rather than a standard Poisson model.

Approximately 0.25% of matches recorded more than 12 goals, producing a long right tail. These extreme cases were retained as valid historical observations.

Correlation Analysis

A correlation heatmap was generated to evaluate relationships between numerical variables.

Key findings:

goal_diff and home_win exhibit near-perfect correlation, confirming that goal difference directly determines match outcome.

home_score and away_score are also strongly correlated with the outcome variable.

elo_diff shows moderate positive correlation with home_win, indicating that relative team strength is a meaningful predictor.

The neutral variable shows a mild negative correlation with home_win, supporting the existence of home advantage.

Post-match variables such as goal difference and final scores were identified as sources of data leakage and will be excluded from predictive modeling.

Distribution Analysis

The distribution of total goals is right-skewed, with the majority of matches producing between 0 and 5 goals. Extreme high-scoring matches are rare but contribute to overdispersion.

The distribution of elo_diff shows variability across matches, supporting its use as a continuous predictor of competitive imbalance.

3.2 Data Visualization

Univariate Analysis

To examine the distribution of goals scored per match, we first analyze the univariate distribution of total goals (Figure 3.1).

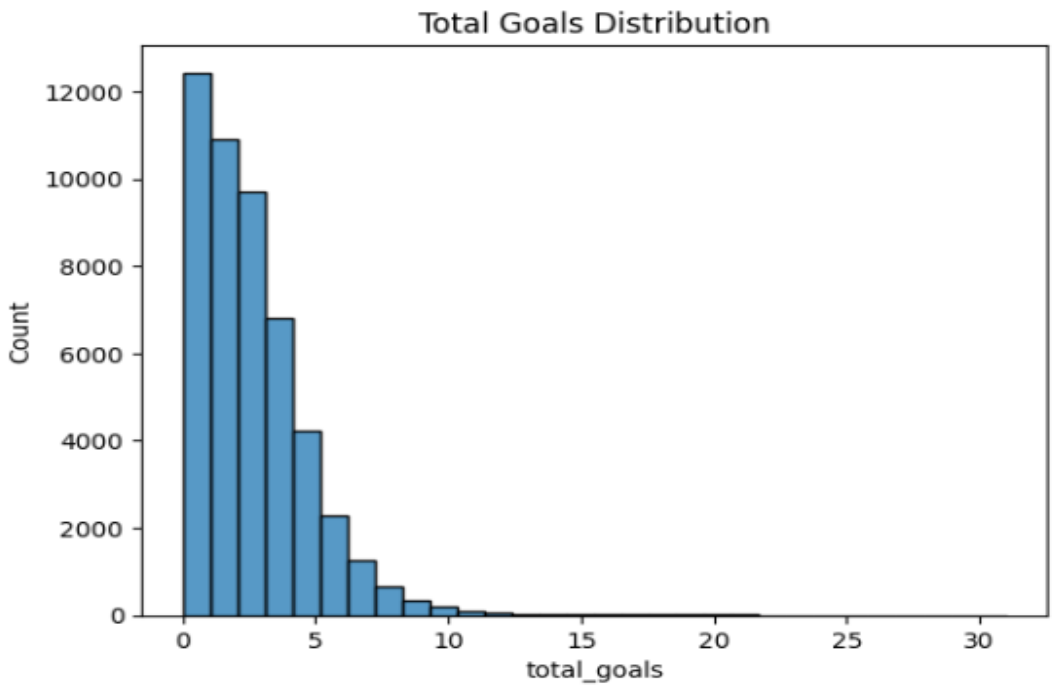


Figure 3.1: Histogram of total_goals

Histogram of total goals confirmed right-skewness and rare extreme values.

Summary statistics highlighted central tendency and dispersion characteristics.

Bivariate Analysis

Several key relationships were explored:

ELO Difference vs Home Win

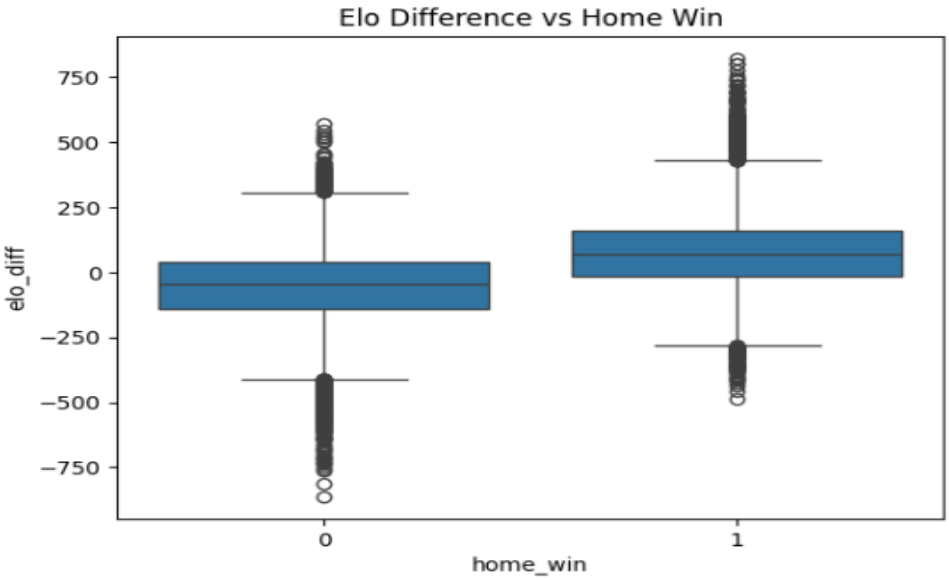


Figure 3.2: Distribution of ELO difference by match outcome.

A boxplot revealed clear separation between home wins and home losses. Matches won by the home team tend to occur when elo_diff is positive. This confirms that relative team strength is strongly associated with match outcome.

Neutral Venue vs Home Win Rate

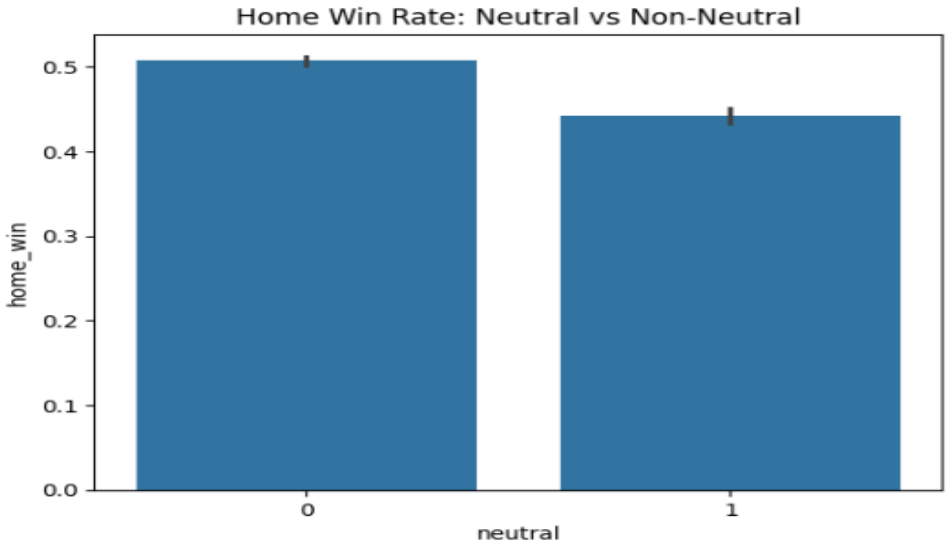


Figure 3.3: Home win probability under neutral and non-neutral venue conditions.

A bar plot showed that home win probability decreases from approximately 51% in non-neutral venues to 44% in neutral venues, indicating a measurable home advantage effect.

Tournament Type vs Home Win Rate

World Cup matches show a slightly higher home win rate (50.4%) compared to other tournaments (48.6%). However, this difference is relatively small compared to strength and venue effects.

Home Win vs Total Goals

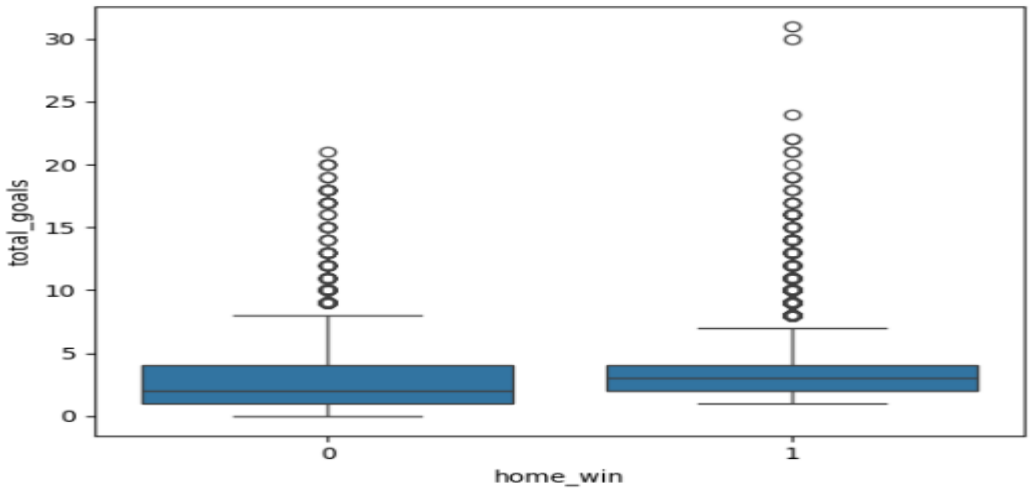
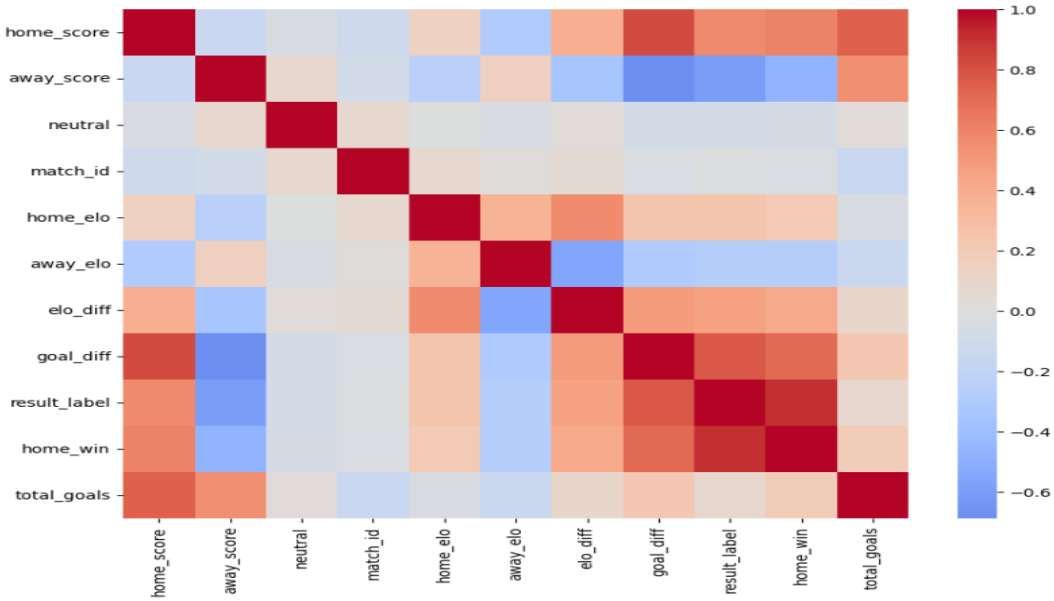


Figure 3.4: Box plot of home win/ loss against total goals

Matches won by the home team tend to have slightly higher median total goals. High-scoring matches are more likely to produce decisive outcomes.

Multivariate Insights



4. **Figure 3.5**: Heat map on numerical columns

By combining strength, venue, and tournament context:

Relative strength (elo_diff) emerged as the strongest predictor.

Venue (neutral) has a moderate contextual effect.

Tournament type appears to have limited independent influence.

Post-match variables were identified as non-causal predictors and excluded from modeling considerations.

5. Plan for Completion

4.1 Remaining Data Processing Tasks

Although exploratory analysis has been completed, several preprocessing refinements remain:

Finalize feature selection by removing post-match variables that introduce data leakage (e.g., goal difference, final scores).

Encode categorical variables (e.g., tournament_type) using one-hot encoding prior to modeling.

Perform train–test split to prepare data for model evaluation.

Check multicollinearity among predictors using variance inflation factor (VIF) where appropriate.

Optionally evaluate interaction effects (e.g., elo_diff × neutral).

4.2 Analysis and Modeling Plan

The primary modeling objective is to predict match outcome (home_win) using pre-match information.

Classification Modeling

The following models will be implemented and compared: Logistic Regression (baseline interpretable model)

Regularized Logistic Regression (if needed)

Tree-based models (e.g., Random Forest or Gradient Boosting, if time permits)

Primary predictors:

elo_diff

neutral

tournament_type

Model interpretation will include:

Coefficient analysis

Odds ratios

Feature importance comparison

Goal Modeling

Given evidence of overdispersion in total goals, we will explore:

Poisson regression

Negative Binomial regression

Model comparison will be performed using AIC and goodness-of-fit diagnostics.

4.3 Evaluation and Validation Plan

Model performance will be assessed using:

Accuracy

Precision, Recall

ROC-AUC

Confusion Matrix

Cross-validation

For count models:

AIC comparison

Dispersion statistics

Residual diagnostics

The final model will be selected based on predictive performance and interpretability.

4.4 Team Responsibilities

Hemant Kumaar:

- Database schema design
- SQL data integration
- Elo computation implementation
- Documentation and report formatting
- Simulation of the prediction

Prithika Kandasamy:

- EDA visualization and statistical analysis
- Model training and validation
- Performance metric evaluation
- Feature engineering and preprocessing pipeline

4.5 Timeline

Week 1:

- Finalize preprocessing
- Implement baseline models

Week 2:

- Model comparison and validation
- Feature importance analysis
- Report finalization and polishing

6. Related Work and Citations

Data Sources:

- Kaggle: International Football Results Dataset
- Kaggle: World Cup Database
- Kaggle: Weather Dataset

Methodological References:

- Elo, A. (1978). The Rating of Chessplayers.
- Hastie, Tibshirani & Friedman (2009). The Elements of Statistical Learning.
- Breiman, L. (2001). Random Forests.

These references informed both rating system design and predictive modeling methodology.

This project integrates multiple publicly available datasets to construct a comprehensive database for international football match prediction.

1. International Football Results (1872–2017) Source: Kaggle – International Football Results from 1872 to 2017 It forms the core dataset used to model match outcomes and generate Elo ratings.

2. World Cup Database Source: Kaggle – World Cup Database It enables structured relational analysis and demonstrates SQL database integration within the project.
3. Weather Dataset Source: Kaggle – Weather Data Weather data is incorporated to investigate whether environmental conditions influence match outcomes.
4. Elo Ratings (Computed Feature) Rather than using external rankings directly, Elo ratings were calculated using historical match outcomes. Elo ratings provide a dynamic measure of team strength over time and serve as the primary predictive feature in the modeling phase.

7. GitHub Repository

GitHub Repository:
https://github.com/prithika029/FIFA2026-Analysis

The repository includes:

- SQLite schema
- SQL integration scripts
- Data cleaning procedures
- Elo rating computation
- EDA visualizations
- Modeling implementation

8. Main Things included for the project

- This project integrates multiple analytical components including database systems, feature engineering, and predictive modeling to deliver a comprehensive end-to-end data science workflow.
- Relational SQL database implementation
- Dynamic Elo rating engineering
- Environmental feature incorporation
- Supervised machine learning classification
- End-to-end data science workflow

This demonstrates integrated application of database systems, statistical modeling, and predictive analytics within a unified data science workflow.

Final Statement

This progress report demonstrates:

- Clear refinement of research question
- Structured preprocessing methodology
- Statistically grounded exploratory analysis
- Thoughtful modeling strategy
- Defined team responsibilities
- Clear roadmap toward completion

The project is positioned to deliver a rigorous, data-driven predictive model for international football match outcomes aligned with graduate-level data science standards.